



**FURTHER EDUCATION AND TRAINING CERTIFICATE:
INFORMATION TECHNOLOGY: SYSTEMS DEVELOPMENT**

ID 78965 LEVEL 4

BLOCK 2 SUMMATIVE ASSESEMENT

FULL NAME & SURNAME	
ID NUMBER:	
NAME OF ASSESSOR	Kabelo P. Matlakala
DATE OF ASSESSMENT	
VENUE	

	ACHIEVED	NOT ACHIEVED
KNOWLEDGE		
SKILLS		

Signature of learner	Signature of Assessor

ASSESSMENT PACK

Please complete the following sections (A) before commencing with this assessment.

Section A Learner Information															
Name:															
Surname:															
School:															
Department:															
Contact no:															
Date:															
ID															

Results:

PRE-ASSESSMENT MEETING CHECKLIST

ACTION	YES/NO	COMMENTS
Set learner at ease; be friendly, polite and professional.		
Explain to the learner and agree on the following issues. 1. The unit standard that will be assessed 2. Date, time, venue and process to be followed during the assessment. 3. Summative assessment tools to be used for the assessment. 4. The assessment plan 5. Purpose of assessment		
Explain to the learner and agree on the role of all involved during the assessment process.		
Identify possible barriers and or disabilities of the learner.		
Explain the meaning and application of RPL.		
Explain, discuss and provide one complete set of the Appeals process documentation.		
Explain to the learner when final results will be available and how feedback will be provided.		
Discuss previous assessment results if applicable.		

I, _____ (initials and surname of learner), DECLARE THE FOLLOWING:

A copy of the unit standard(s) involved has been given to me prior to this meeting. I know I will be assessed against the criteria, which have been set to the applicable unit standards. The criteria have been discussed with me, and the procedures and purpose of the assessment has been clearly explained to me.

I am well aware of the venue, date and time that I will be assessed. I consider the period of time given to me to prepare myself for the assessment to be fair.

I understand clearly that I have the right to appeal against any decision made by the assessor during the assessment of the evidence provided by me, and that I have free access to the appeals procedures attached to this assessment pack. I understand that I have the right to be accompanied by another person during all procedures, and that I have free access to the Training Division of Health and Safety Procedures- filed at the offices.

Signature of learner	Date

Assessment Instruments

TAKE NOTE

- The instruments in this pack are all for summative assessment.
- Use these together with the formative assessment activities in the learner workbook.
- Both formative (workbook) and summative assessments must be kept in the learner's Portfolio of Evidence (PoE).

A number of the assessment instruments contained in this assessment are workplace knowledge based questions. This means that you will arrange with the learner, a time that is suitable, during which the learner will complete each questions.

Instructions to Learner

1. Answer ALL questions in ALL three sections.
2. For Section A (MCQ): circle or tick ONE answer per question.
3. For Section B (short answer): write in clear, complete sentences or select from the provided options.
4. Write legibly. Illegible answers will not be marked.
5. No calculators, notes, or electronic devices permitted.

NOTE: Section 1 covers programming principles (40 marks), Section 2 covers web scripting (40 marks), and Section 3 covers internet technologies (20 marks). Manage your time accordingly.

Section	Module	Total	Mark Awarded
1	14910 — Apply Principles of Computer Programming	40	
2	14933 — Create Web Applications Using Scripting	40	
3	14930 — Developing Software for Internet Technologies	20	
	TOTAL	100	

Section 1 · Module 14910 | Applying Programming Principles**[40 marks]****Section A: Multiple Choice Questions [20 marks — 2 marks each]**

Circle or tick **ONE** correct answer for each question.

1. Which data type is most appropriate to store a student's age?

A) float
B) boolean
C) integer
D) string

2. A programmer writes the pseudocode below. What will it print when score = 72?

```
IF score ≥ 75 THEN
    PRINT "Distinction"
ELSE IF score ≥ 60 THEN
    PRINT "Pass"
ELSE
    PRINT "Fail"
END IF
```

- A) Distinction
B) Pass
C) Fail
D) Nothing — the code contains an error
3. Which loop structure is best suited when the number of repetitions is known in advance?
- A) WHILE loop
B) FOR loop
C) DO-WHILE loop
D) REPEAT-UNTIL loop
4. A student's code runs without crashing but always calculates the wrong total. What type of error is this?
- A) Syntax error
B) Runtime error
C) Logic error
D) Compilation error

5. Which of the following is a logical operator used to combine two conditions?
- A) +
 - B) >=
 - C) AND
 - D) ==
6. What is the output of: `PRINT 5 + 3 * 2`?
- A) 16
 - B) 11
 - C) 13
 - D) 26
7. What does `#` mean in Python?
- A) It is a variable
 - B) It is a comment
 - C) It is a loop
 - D) It is an error symbol
8. Which number system is used by computers and consists only of 0s and 1s?
- A) Binary
 - B) Decimal
 - C) Hexadecimal
 - D) Octal
9. Which statement about functions / subroutines is true?
- A) They can only be called once
 - B) They help avoid code duplication
 - C) They must always return a value
 - D) They cannot accept parameters
10. What happens when you divide an integer by zero in most programming languages?
- A) It returns zero
 - B) It returns infinity
 - C) It causes a runtime error
 - D) It returns the integer itself

Section B: Short-Answer Questions [20 marks]

B1. Build the pseudocode by selecting the best line in sequence for each step. (10 marks)

The pseudocode should:

- (1) Loop from 1 to 5,
- (2) Add each number to a running total,
- (3) Display the result.

■ SET total = 0

■ FOR total = 1 TO 5

■ INPUT total

■ FOR i = 1 TO 5

■ PRINT total

■ END FOR

■ WHILE i < 5

■ END WHILE

■ SET i = total + 1

■ OUTPUT i

Line 1 — your answer:	
Line 2 — your answer:	
Line 3 — your answer:	
Line 4 — your answer:	
Line 5 — your answer:	

Final Pseudocode(optional):

B2. Error detection task.

(4 marks)

The pseudocode below is intended to print PASS when average ≥ 50 .

```
IF average > 50 THEN
  PRINT "PASS"
ELSE
  PRINT "FAIL"
END IF
```

a) What is wrong with the logic? (2 marks)**Options:**

- ☐ The condition uses > instead of $\geq$, so a student with exactly 50 will incorrectly get FAIL
- ☐ The ELSE branch is missing
- ☐ The PRINT statements are swapped
- ☐ There is no error — the logic is correct

b) Rewrite the correct condition. (2 marks)**Options:**

- ☐ IF average ≥ 50 THEN
- ☐ IF average > 50 THEN
- ☐ IF average = 50 THEN
- ☐ IF average < 50 THEN

Final Pseudocode(optional):

B3. A student writes a program to calculate totals but gets incorrect results and performs desk-checking.

1) What is desk-checking? (2 mark)

- a) Manually tracing program logic using sample data without running the code
- b) Running the program on a computer to find errors
- c) Asking a colleague to read through the code
- d) Compiling the code to check for syntax errors

2) How would it help? Choose ONE error it could detect. (2 marks)

- a) Incorrect loop counter causing wrong number of iterations
- b) A missing semicolon that stops compilation
- c) A network timeout during data submission
- d) A missing import statement

B4. Code tracing and debugging. Study the Python code below: (4 marks)

```
total = 0
for i in range(2): # repeats 2 times
    total = total + 10
print(total)
```

1) Trace the code? (2 mark)

What value will be printed?

Answer:	
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2) Identify and fix the error (2 marks)

The program should add 10 five times.

- a) What is the mistake in the code?
 - a. The print statement is missing
 - b. The value 10 is incorrect
 - c. The loop runs 2 times instead of 5
 - d. The total is not initialised

b) Rewrite the incorrect line to fix the program.

Correct line:	
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Section 2 · Module 14933 | Create Web Applications Using Scripting [40 marks]**Section A: Multiple Choice Questions [20 marks — 2 marks each]**

Circle or tick **ONE** correct answer for each question.

1. Which HTML element defines the main content area visible in a browser?
 - A) `<head>`
 - B) `<meta>`
 - C) `<body>`
 - D) `<!DOCTYPE>`
2. A developer wants to style ALL `<p>` elements on a page to have red text. Which CSS rule is correct?
 - A) `#p { color: red; }`
 - B) `p { color: red; }`
 - C) `.p { color: red; }`
 - D) `p { text-color: red; }`
3. Which JavaScript statement correctly retrieves an HTML element with the id `"studentName"`?
 - A) `document.getElement("studentName")`
 - B) `document.querySelector("#studentName")`
 - C) `document.getElementById("studentName")`
 - D) Both B and C are correct
4. Which JavaScript keyword is used to declare a variable that cannot be reassigned?
 - A) `var`
 - B) `let`
 - C) `const`
 - D) `static`
5. What happens when a user clicks `<input type="submit">` in a form?
 - A) The page reloads with no action
 - B) The form is submitted or handled by a submit event
 - C) The input field is cleared automatically
 - D) A new tab opens

6. Which CSS property changes the text colour of an element?
- A) font-size
 - B) color
 - C) background-color
 - D) border
7. Which statement best describes web-based multimedia?
- A) Text-only web pages
 - B) Websites combining media types with interaction
 - C) Image-only websites
 - D) Audio-only websites
8. Which HTML tag creates a hyperlink?
- A) <link>
 - B) <a>
 - C) <href>
 - D) <url>
9. What does CSS stand for?
- A) Computer Style Sheets
 - B) Cascading Style Sheets
 - C) Creative Style Sheets
 - D) Colour Style System
10. Which of the following is the BEST way to improve web page performance and loading speed?
- A) Use large, high-quality images without optimisation
 - B) Load all videos fully before the page displays
 - C) Use optimised images and thumbnails for large content
 - D) Add more animations and media to improve appearance

Section B: Short-Answer Questions [20 marks]

B1. A learner creates a simple form. When they click submit, the page refreshes and any message disappears. (6 marks)

a) Why does this happen?

Choose ONE:

- The browser reloads the page automatically when a form is submitted
- The CSS styling is incorrect
- The image on the page is too large
- The HTML page is missing a title

Answer:

b) How can `preventDefault()` fix this?

Choose ONE:

- Stop the page from refreshing when the form is submitted
- Add more images to the page
- Change the background colour
- Add another HTML page

Answer:

c) Where should this fix be applied?

Choose ONE:

- In the JavaScript that runs when the form is submitted
- In the CSS file
- Inside the image tag
- In the page title

Answer:

B2. Design a simple two-page learner support web app.

(6 marks)

Describe Page 1, Page 2, and the navigation flow between them.

a) **Page 1** — purpose and main features**Page 1 options:**

- **Home / Login page** — form with username and password fields, submit button, link to register
- **About Us page** — static company information only
- **Database admin panel only**
- **A blank placeholder page**

Answer:

b) **Page 2** — purpose and main features**Page 2 options:**

- **Support request page** — form with name, email, message fields, submit button, and confirmation message
- **A copy of page 1**
- **A CSS stylesheet only**
- **A server-side database configuration file**

Answer:

c) **Navigation flow between the pages****Navigation options:**

- User clicks "Get support" link on page 1 → navigates to page 2; page 2 has a "Back to home" link
- **No links** — user types the URL manually each time
- **Navigation is handled by the server only**
- **Both pages are on the same URL**

Answer:

B3. Form validation task. For each field below, select one practical validation rule. (4 marks)

Validation rule options:

- 1. Must not be empty and contain letters only
- 2. Must be digits only and exactly 10 characters
- 3. Must be at least 10 characters long
- 4. Must be a future date only
- 5. Must contain @ symbol and a valid domain
- 6. Can accept any value

Field	Answer
Name	
Email	
Contact-Number	
Message	

B4. Identify and explain TWO ways that client-side scripting improves the user experience in form-based pages (4 marks)

Way 1 options:

- Instant validation feedback without a page reload
- Disabling the submit button until all fields are valid
- Storing form data permanently in a server database
- Making the server process requests faster

Answer:	
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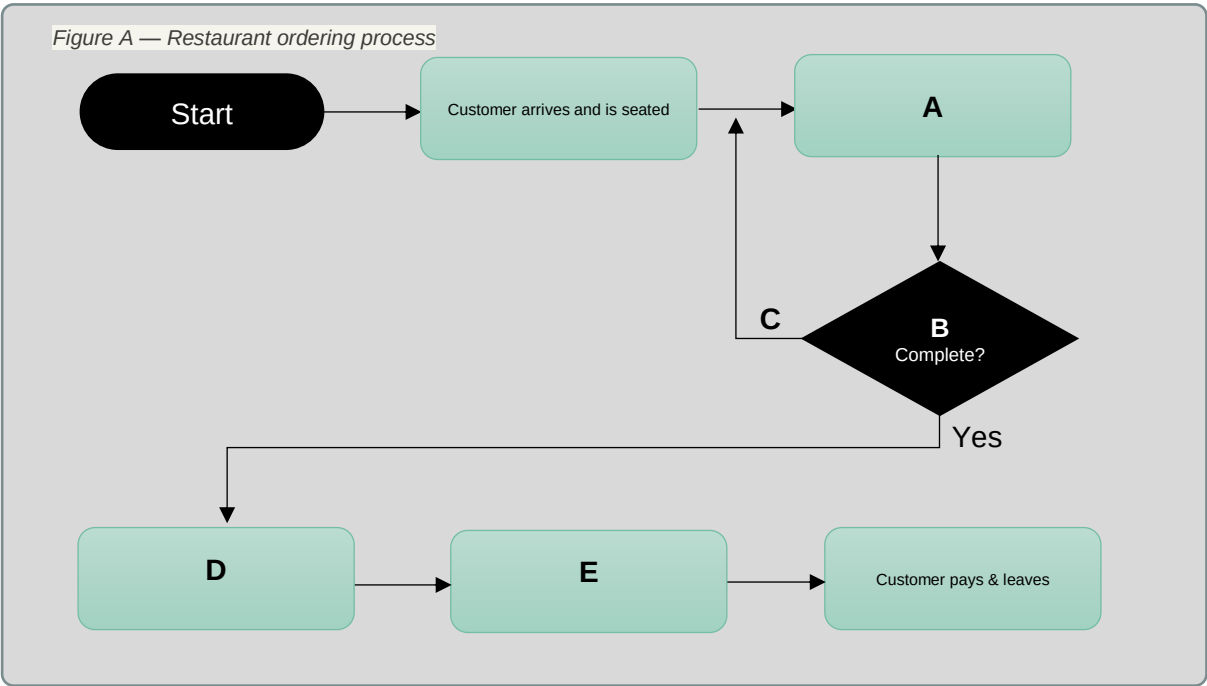
Way 2 options:

- Auto-formatting input fields (e.g. phone numbers)
- Hiding all error messages from the user
- Replacing the need for a back-end server entirely
- Encrypting all data before it reaches the browser

Answer:	
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B2. Study the diagram below (Figure A) and answer the questions that follow. (6 marks)

This diagram represents a restaurant ordering process.



- a) What type of diagram is shown above?
1. Entity Relationship Diagram
 2. Flowchart
 3. Data Flow Diagram
 4. Gantt Chart
- b) Complete labels A–E using the pool below.

Label pool:

- Order
- Waiter takes the order
- Meal served to customer
- Kitchen prepares the meal
- No

Label	Choose the correct step
Label A	
Label B	
Label C	
Label D	
Label E	

Section 3 · Module 14930 | Developing Software for Internet Technologies [20marks]**Section A: Multiple Choice Questions [20 marks — 2 marks each]**

Circle or tick *ONE* correct answer for each question.

1. In a client-server model, which device sends a request for resources or services?
A) Server
B) Router
C) Client
D) Hub

2. A learner types `https://www.cetconnect.co.za/login` into a browser. What does https represent?
A) The domain name
B) The protocol used to transfer data securely
C) The path to the file on the server
D) The port number used by the server

3. Which protocol is used to send email from a client to a mail server?
A) FTP
B) POP3
C) IMAP
D) SMTP

4. What is the purpose of DNS (Domain Name System)?
A) It encrypts web traffic between the client and server
B) It translates human-readable domain names into IP addresses
C) It stores web page files on the hosting server
D) It controls which users can access a website

5. Which of the following best describes a static web page?
A) A page that changes content based on user interaction or a database
B) A page whose content is fixed in the HTML file and looks the same for every visitor
C) A page that requires a login before displaying content
D) A page rendered entirely by JavaScript on the client side

6. Which protocol is used for secure file transfer over the internet?
- A) HTTP
 - B) FTP
 - C) SFTP
 - D) SMTP
7. What is the main purpose of a web server?
- A) To create web pages
 - B) To store and serve web content to clients
 - C) To connect to databases
 - D) To encrypt internet traffic
8. Which of the following is NOT a valid HTTP status code?
- A) 200 OK
 - B) 404 Not Found
 - C) 500 Internal Server Error
 - D) 600 Page Reload
9. What does URL stand for?
- A) Universal Resource Locator
 - B) Uniform Resource Locator
 - C) United Resource Link
 - D) Unique Resource Locator
10. Which component is responsible for routing internet traffic?
- A) Web browser
 - B) Router
 - C) Database
 - D) Firewall

Section B: Short-Answer Questions [9 marks]

B1. For each scenario, classify it as Static or Dynamic and give one short reason. (5 marks)

Reason pool:

- a) Content is fixed in the HTML file and never changes**
- b) Content changes per user based on database queries**
- c) JavaScript modifies the DOM in response to user actions**
- d) The page is styled with CSS**
- e) The server sends the same HTML to every visitor**

Scenario	Static / Dynamic	Select one reason
About Us page with fixed content		
Learner portal showing each user's own marks		
A page where JavaScript changes text after button click		

ASSESSOR & MODERATOR SIGN-OFF

Assessor Name: Kabelo Matlakala

Assessor Signature:

Date Assessed:

Moderator Name:

Moderator Signature:

Date Moderated:

ETQA / SDF Reference:

Notes: